

Models for unbounded count data

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Outline

- ▶ Models for count data
- ▶ Residual diagnostics in GLMs
- ▶ Other useful models

Questions about yesterday?



The binomial GLM

Response data: r the number of successes in N trials

Predictor variables: x_i albeit continuous and/or categorical

Parameters: probability of success p_i in trial i

Goal: estimate π_i for each observation

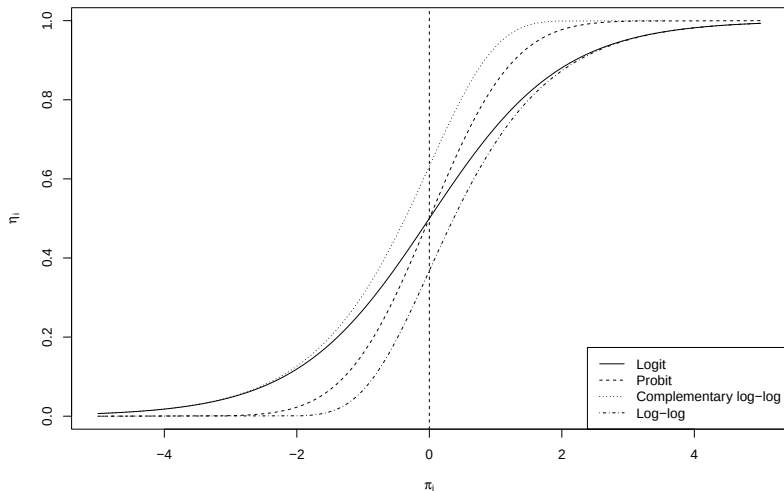
Binomial GLM use

- ▶ When a linear regression is not appropriate :)
- ▶ For binary data or counts of successes/failures

Common examples

- ▶ OME signal identification
- ▶ cancer rates
- ▶ Predicting species' distributions
- ▶ Number of germinated plant seeds
- ▶ Prevalence of disease in a population
- ▶ Probability of observing a behavior
- ▶ Proportion of orchids 🙄

Binomial link functions (2)



Typical count cases

- ▶ Number of caught fish
- ▶ Number of deaths due to lung cancer or other diseases
- ▶ Seizure counts
- ▶ Times a behavior is expressed
- ▶ Number of pigeons in a city
- ▶ Number of Bigfoot reports
- ▶ Number of wrongful convictions
- ▶ Number of stars in the night sky

The Poisson GLM

Response data: k_i the count

Predictor variables: x_i albeit continuous and/or categorical

Parameters: mean λ

Goal: estimate λ_i for each observation

The Poisson distribution

$$\mathcal{L}(y_i; \Theta) = \exp\{y_i \log(\lambda_i) - \lambda_i - \log(y_i!)\} \quad (1)$$



The Poisson parameter λ is the mean of the counting process

Is Poisson regression in the EF?

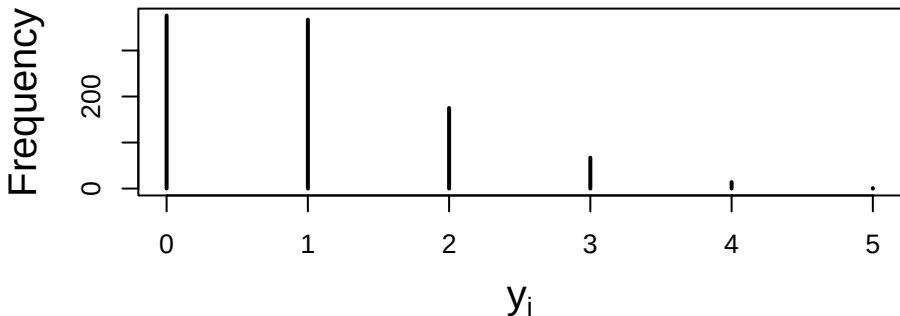
$$\mathcal{L}(y_i; \Theta) = \exp \left\{ \frac{y_i \log(\lambda) + \log(\lambda)}{1} + \log(y_i!) \right\} \quad (2)$$

All GLMs can be formulated as:

$$\mathcal{L}(y_i; \Theta) = \exp \left\{ \frac{y_i \eta_i - b(\eta_i)}{a(\phi)} + c(y_i, \phi) \right\} \quad (3)$$

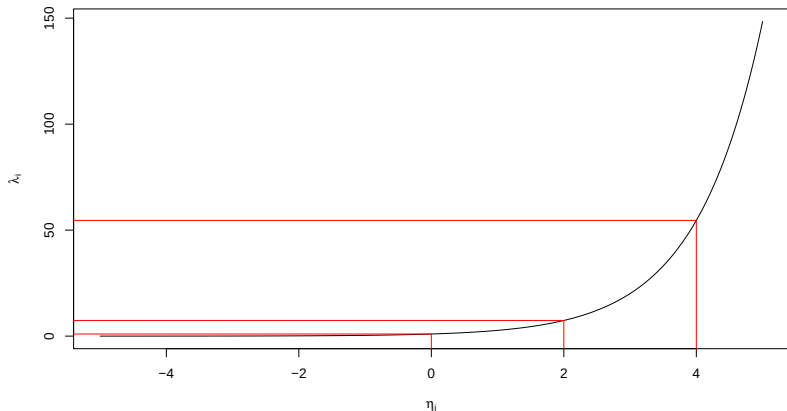
The Poisson distribution visually

$$\lambda = 1$$



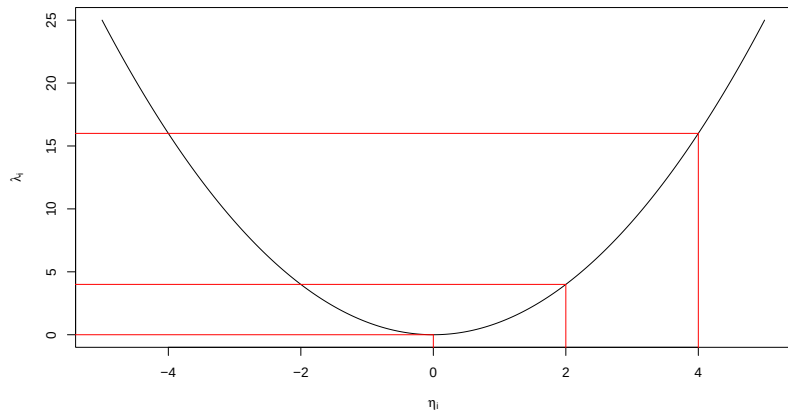
Log-link function

So log is the canonical link. This looks like:



square root-link function

An alternative is the square root-link function $\lambda_i = (\alpha + x_i\beta)^2$



Poisson assumptions

- ▶ An event can occur $0 \dots \infty$ times
- ▶ Events are independent
- ▶ Events cannot occur simultaneously
- ▶ Variance equals the mean
- ▶ The rate of events is constant

The rate of events

Counts are usually collected over time or space:

- ▶ The amount of fish we catch in an hour
- ▶ The number of deaths on a population of 10.000
- ▶ The number of seizures a patient has during the night
- ▶ The number of times a behavior is expressed during a treatment
- ▶ Number of pigeons in a city the size of Los Angeles
- ▶ The number of bigfoot reports collected in a small forest, yesterday, by 3 people
- ▶ Number of wrongful convictions in Germany last year
- ▶ Number of stars in the night sky

The Poisson distribution: rates

Alternatively we can write:

$$\mathcal{L}(y_i; \Theta) = \exp\{y_i \log(rt) - rt - \log(y_i!)\} \quad (4)$$

so, $\lambda = rt$

- ▶ r is the rate at which counts occur, per time period t
- ▶ we can instead write $\lambda = t \exp(\eta) = \exp\{\eta + \log(t)\}$
- ▶ $\log(t)$ is called an **offset**

Example: going out fishing

- ▶ On average we catch 1 fish in 20 minutes
 $\lambda = 1 = \exp\{-2.99 + \log(20)\}$
- ▶ If we go fishing for an hour we catch $\exp(-2.99) * 60 = 3$ fish
- ▶ If we go fishing for one minute we catch $\exp(-2.99) = \frac{1}{20}$ fish
- ▶ Here, $r = \exp(-2.99)$ and t is the time we want to spend fishing
- ▶ We can also find the amount of time we need to spend to catch 5 fish
 - ▶ $\exp\{-2.99 + \log(t)\} = 5$, so $t = \frac{5}{\exp(-2.99)} = 100$ minutes

Log-linear regression

Log-linear regression is a class of models that uses the log-link function:

$$\begin{aligned}\log\{\mathbb{E}(y_i|x_i)\} &= \eta_i = \alpha + x_i\beta \\ \mathbb{E}(y_i|x_i) &= \lambda_i = \exp(\alpha + x_i\beta)\end{aligned}\tag{5}$$

Log-linear regression is commonly used to analyse count data

Log-linear regression

Log-linear regression is a “multiplicative” model

$$\begin{aligned}
 \lambda_i &= \exp(\alpha + x_i\beta) \\
 &= \exp(\alpha) \exp(x_i\beta)
 \end{aligned}
 \tag{6}$$

Log-linear regression

Log-linear regression is a “multiplicative” model

$$\begin{aligned}\lambda_i &= \exp(\alpha + x_i\beta) \\ &= \exp(\alpha) \exp(x_i\beta)\end{aligned}\tag{6}$$

A unit increase in x_i scales λ_i by $\exp(\beta)$

So, when $\exp(\beta) = \frac{1}{2}$, $\exp(\alpha)$ halves for every unit of x_i

So, when $\exp(\beta) = 2$, $\exp(\alpha)$ doubles for every unit of x_i

Log-linear regression

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So, when $\exp(\beta) = 2$, $\exp(\alpha)$ doubles for every unit of x_i

Of course, this is more involved with multiple predictors

Example of a multiplicative process

Say that we have the model:

$$\log(\lambda) = \alpha + x_i\beta \quad (7)$$

- ▶ with $\alpha = -2.99$ and $\beta = \log(2) \approx 0.693$
- ▶ x_i is either 0 or 1: either I was fishing or you were
- ▶ $\exp(-2.99) = 0.05$ the average number of fish I caught in the time I spent fishing
- ▶ $\exp(-2.99 + \log 2) = \exp(-2.99) * 2 = 0.1$ the average number of fish you caught
- ▶ So, you caught twice as many fish! I am not very good at fishing

Example: campus crime

Count of violent crimes for an academic year



Figure 1: freepik.com



Figure 2: campussecuritytoday.com

Campus crime: the data

Data via Legler and Roback (2021)

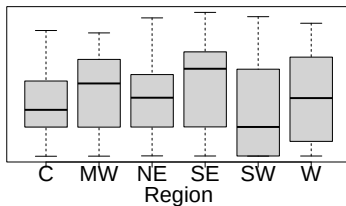
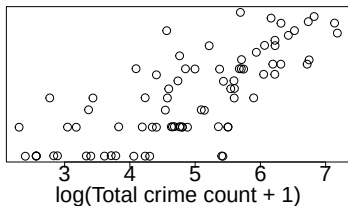
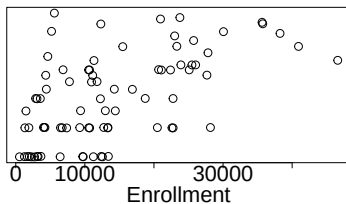
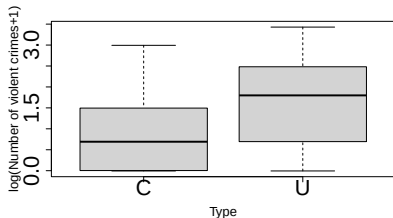
- ▶ 81 observations
- ▶ Number of violent crimes, Total number of crimes, Number of property crimes
- ▶ Student enrollment
- ▶ Type (University or College)
- ▶ Region of the country

num_viol	total_crime	num_prop	Enrollment	type	region
30	296	266	5590	U	SE
0	10	10	540	C	SE
23	1256	1233	35747	U	W
1	211	210	28176	C	W
1	117	116	10568	U	SW
0	29	29	3127	U	SW
7	291	284	20675	U	W

How could these data be analysed with a binomial regression?

num_viol	total_crime	num_prop	Enrollment	type	region
30	296	266	5590	U	SE
0	10	10	540	C	SE
23	1256	1233	35747	U	W
1	211	210	28176	C	W
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0	29	29	3127	U	SW
7	291	284	20675	U	W

What is the relationship between violent crimes and school variables?

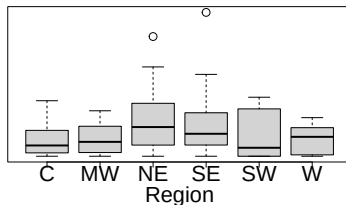
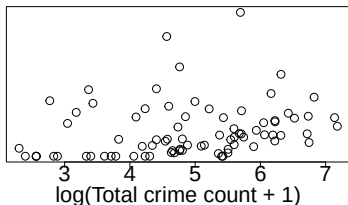
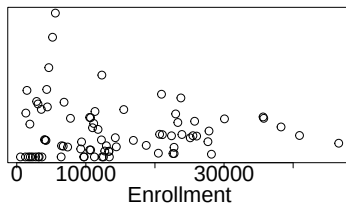
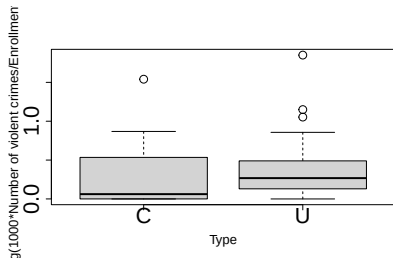


Campus crime: fit the model

```
model <- glm(num_viol ~ region+type,
              family = "poisson", data = campus)
```

What issue can we identify for this model?

What is the relationship between violent crimes per 1000 enrolled and school variables?



Campus crime: fit the model

```
modelo <- glm(num_viol ~ region + type
               + offset(log(Enrollment)),
               family = "poisson", data = campus)
```

##	No.offset	Offset
## (Intercept)	0.34	-8.509
## regionMW	0.49	0.099
## regionNE	0.52	0.781
## regionSE	0.87	0.877
## regionSW	0.25	0.503
## regionW	0.74	0.273
## typeU	1.24	0.340

Campus crime: per 1000 enrolled

```

model1000 <- glm(num_viol ~ region + type
                  + offset(log(Enrollment/1000)),
                  family = "poisson", data = campus)
  
```

$$\exp(\alpha_2) * 1000 = \exp(\alpha_3) \quad \exp(\alpha_2) = \exp(\alpha_3)/1000$$

$$\exp(\alpha_2) = \exp\{\alpha_3 - \log(1000)\}$$

Campus crime: per 1000 enrolled

##	No.offset	Offset	Per thousand
## (Intercept)	0.34	-8.509	-1.602
## regionMW	0.49	0.099	0.099
## regionNE	0.52	0.781	0.781
## regionSE	0.87	0.877	0.877
## regionSW	0.25	0.503	0.503
## regionW	0.74	0.273	0.273
## typeU	1.24	0.340	0.340

Do we have a good model?

More on this after the break

- ▶ Overdispersion or underdispersion
- ▶ Zero-inflation

Overdispersion

Our assumption: $\lambda = \text{var}(\mathbf{y})$

Reality: $\lambda \geq \text{var}(\mathbf{y})$

- ▶ Mean = variance
- ▶ If there is more variation, this assumption fails
- ▶ Consequences: CIs underestimate, biased parameter estimates, inflation in model selection

For our example: many females have few satellites, but some females have very many.

Underdispersion

Our assumption: $\lambda = \text{var}(\mathbf{y})$

Reality: $\lambda \leq \text{var}(\mathbf{y})$

Considerably less common than overdispersion.

Detecting overdispersion

- ▶ Residual diagnostics
- ▶ $D(\mathbf{y}; \hat{\boldsymbol{\mu}})/(n - k)$: should be close to 1
- ▶ `performance::check_overdispersion` (relies on asymptotics)
- ▶ Simulation (later today)

Dealing with dispersion: options

- ▶ Correct for it (calculate dispersion)
- ▶ Fit a different model
 - ▶ Negative binomial (overdispersion, MASS package)
 - ▶ Conway-Maxwell Poisson (over- and underdispersion.)
 - ▶ Generalized Poisson(over- and underdispersion)
 - ▶ Quasi-likelihood models
 - ▶ Mixed models (not covered here)

Quasi-likelihood models

Introduced by [Wedderburn \(1974\)](#)

- ▶ No “real” likelihood is specified for the data
- ▶ Means no AIC, but deviance exists
- ▶ Largely defined by its variance function

For Poisson responses: does not correct the parameter estimates

Negative-binomial

$$\mathcal{L}(y_i; \Theta) = \frac{\Gamma(y_i + \phi)}{\Gamma(\phi)y_i!} \left(\frac{\phi}{\mu_i + \phi} \right)^\phi \left(\frac{\mu_i}{\mu_i + \phi} \right)^{y_i} \quad (8)$$

- ▶ $\text{var}(\mathbf{y}) = \boldsymbol{\mu} + \frac{\boldsymbol{\mu}^2}{\phi}$
- ▶ For large ϕ Poisson!
- ▶ Requires more data/information due to extra parameter

Is negative-binomial regression in the EF?

$$\mathcal{L}(y_i; \Theta) = \exp\left[\frac{y_i \log\left\{\frac{\mu_i}{\mu_i + \phi}\right\} - \phi \log\left\{\frac{\mu_i + \phi}{\phi}\right\}}{1} + \log\{\Gamma(y_i + \phi)\} - \log\{\Gamma(\phi)\} - \log(y_i!)\right] \quad (9)$$

All GLMs can be formulated as:

$$\mathcal{L}(y_i; \Theta) = \exp\left\{\frac{y_i \eta_i - b(\eta_i)}{a(\phi)} + c(y_i, \phi)\right\} \quad (10)$$

Example: hurricanes

Deaths due to hurricanes



Figure 2: Satellite image of a hurricane

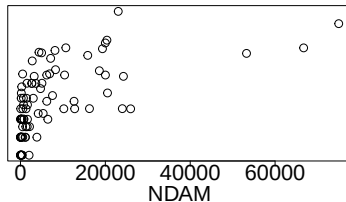
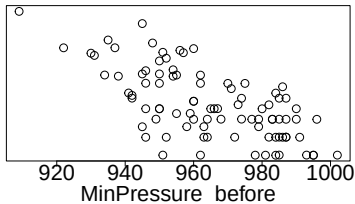
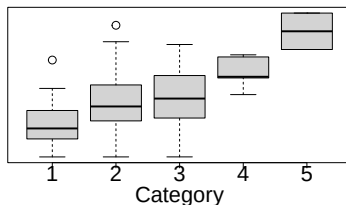
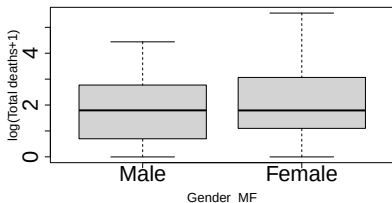
Hurricanes: the data

Data from Jung et al. (2014)

- ▶ 94 observations
- ▶ Year, name, Binary name categorization (male, female), Masulinity-Femininity score, strength of the hurricane, prior air pressure
- ▶ Excluded two outliers

Year	Name	MasFem	MinPressure_before	Gender_MF	Category	alldeaths	NDAM	Elapsed_Yrs
1950	Easy	6.777778	958	Female	3	2	1590	63
1950	King	1.388889	955	Male	3	4	5350	63
1952	Able	3.833333	985	Male	1	3	150	61
1953	Barbara	9.833333	987	Female	1	1	58	60
1953	Florence	8.333333	985	Female	1	0	15	60
1954	Carol	8.111111	960	Female	3	60	19321	59
1954	Edna	8.555556	954	Female	3	20	3230	59

Are female hurricanes more deadly than male hurricanes?



Hurricanes: fit the model

```
modelp <- glm(alldeaths ~ Gender_MF + MinPressure_before,
               family = "poisson", data = hurricanes)
```

Hurricanes: interpreting parameters

##		Estimate	Std. Error	z value	Pr(> z)
##	(Intercept)	55.000	1.2000	46.0	0.0e+00
##	Gender_MFFemale	0.270	0.0570	4.8	1.7e-06
##	MinPressure_before	-0.055	0.0013	-43.0	0.0e+00

How do we interpret the intercept?

Hurricanes: interpreting parameters

```
##                                Estimate Std. Error z value      Pr(>|z|)
## (Intercept)                   2.100      0.0550    38.0 9.881313e-
324
## Gender_MFFemale               0.270      0.0570     4.8 1.700000e-
06
## MinPressure_beforeC          -0.055      0.0013     -
43.0 0.000000e+00
```

Average prior air pressure: 964.49 knots

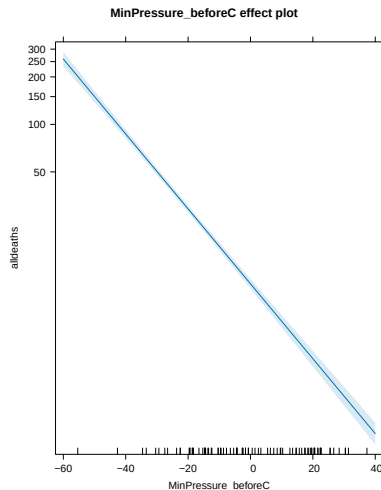
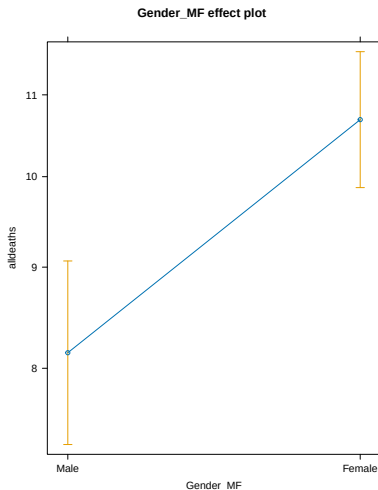
How do we interpret the intercept?
And its standard error?

Prior air pressure centered

Hurricanes: interpreting parameters on the response scale

- ▶ (Intercept) = Male-named hurricanes, prior air pressure ≈ 965
- ▶ Female-named hurricanes: $\exp(0.27) = 1.3$, so 30% more deadly

Hurricanes: visual interpretation



Hurricanes: checking overdispersion

```
performance::check_overdispersion(modelp1)
```

```
## # Overdispersion test
##
##      dispersion ratio =    18.974
##  Pearson's Chi-Squared = 1650.771
##      p-value =    < 0.001
```

```
## Overdispersion detected.
```


Hurricanes: Negative-binomial

```
modelnb <- MASS::glm.nb(alldeaths ~ Gender_MF + MinPressure_beforeC
```

and compare the models:

```
AIC(modelp1, modelnb)
```

```
##          df      AIC
## modelp1   3 1639.9581
## modelnb   4  612.1051
```

Hurricanes crabs: comparing estimates

##	Poisson estimate	NB estimate	Poisson SE	NB SE
## (Intercept)	2.100	2.200	0.0550	0.2000
## Gender_MFFemale	0.270	0.047	0.0570	0.2500
## MinPressure_beforeC	-0.055	-0.056	0.0013	0.0061

- ▶ SEs have increased
- ▶ Coefficients have changed
- ▶ **Female and male-named hurricanes are equally deadly**
- ▶ Effect of pressure has remained

Count distributions

- ▶ Poisson
- ▶ Negative binomial (two types, with dispersion)
- ▶ Conway-Maxwell Poisson (with dispersion)
- ▶ Generalized Poisson (with dispersion)
- ▶ Skellam distribution (difference of counts)
- ▶ Binomial distribution (counts with a maximum)
- ▶ Truncated distributions (e.g., without zeros)
- ▶ Quasi-likelihood models

Summary

- ▶ Counts are analysed with log-linear models
- ▶ The collection effort of counts needs to be considered (offset)
- ▶ When the Poisson assumption is violated, we change to another count distribution